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Topic : Containerization and Virtualization : Key Technical Terms :

Virtualization and Containerization - Key Technical Terms :

Term	Technical Meaning	Easy Meaning
Virtual Machine (VM)	Software-based computer that runs on physical hardware via a hypervisor with its own OS.	Computer inside another computer.
Virtualization	Technology that creates virtual versions of servers, OS, or storage.	Splitting one computer into many virtual computers.
Infrastructure Layer	Physical hardware providing CPU, memory, storage, and networking.	Actual hardware like CPU, RAM, disk.
Hypervisor	Software that creates and manages virtual machines by allocating hardware resources.	Manager that divides hardware for VMs.
Virtual Machine Monitor (VMM)	Another name for hypervisor that monitors VM resource usage.	Brain controlling virtual machines.

Hardware Abstraction	Creating virtual versions of hardware components for VMs.	Fake hardware for VMs.
Isolation	Ensures VMs or containers run independently.	Each runs in its own box.
Resource Scheduling	Process of allocating CPU, RAM, and storage among VMs.	Decides who gets CPU/RAM.
Emulation	Software imitation of hardware devices.	Software acting like real hardware.
Guest OS	Operating system running inside a virtual machine.	OS inside the VM.
Kernel	Core part of OS managing CPU, memory, and devices.	Brain of the OS.
Device Drivers	Software enabling OS to communicate with hardware.	Programs that let OS talk to hardware.
System Libraries	Reusable code used by applications.	Ready-made helper code.
File System	Structure used by OS to organize and store data.	Method to store files in folders.
System Services	Background processes providing system functionality.	Programs running quietly in background.

Binaries	Executable program files that OS can run directly.	Files that run programs.
Libraries	Reusable code modules shared by programs.	Toolbox of code used by apps.
Dependencies	External software packages required by applications.	Extra tools an app needs.
Container	Lightweight package containing app and dependencies sharing host kernel.	Small box with app and tools.
Containerization	Packaging apps with dependencies into containers.	Putting an app into a portable box.
Host Operating System	Main operating system installed on the physical machine.	Primary OS of the computer.
Container Engine	Software that creates and manages containers (e.g., Docker).	Tool that runs containers.
Container Runtime	Component that starts and stops containers.	Part that actually runs containers.
Container Image	Template containing app, libraries, and configuration.	Snapshot of app ready to run.
Docker Hub	Online registry storing container images.	Library of container images.

Namespaces	Linux feature isolating processes in containers.	Private space for each container.
Control Groups (cgroups)	Linux feature limiting CPU, RAM usage for containers.	Controls resource usage.
Application Layer	Top layer where the actual application runs.	Real program you want to run.
Configuration Files	Files containing application settings.	Files telling program how to behave.
Environment Variables	Dynamic configuration values used by applications.	Settings like ports or passwords.